Journal of Neurotrauma
40:2330–2340 (November 2023)
© Mary Ann Liebert, Inc.
DOI: 10.1089/neu.2022.0043

Journal of Neurotrauma

Open camera or QR reader and scan code to access this article and other resources online.

ORIGINAL ARTICLE CLINICAL STUDIES

Examining Whether Loss of Consciousness Is Associated With Worse Performance on the SCAT5 and Slower Clinical Recovery After Concussion in Professional Athletes

Grant L. Iverson,[1–4] Charles E. Gaudet,[1,3] Alicia Kissinger-Knox,[1–3] and Andrew J. Gardner[5,*]

Downloaded by Massachusetts Institute of Technology from www.liebertpub.com at 11/21/23. For personal use only.

Abstract

Video surveillance has almost universally been employed by professional sports to identify signs of concussion during competition. This study examined associations between video-identified possible loss of consciousness (LOC), acute concussion evaluation findings, and recovery time in concussed professional rugby league players. Medical personnel and sideline video operators identified head impact events sustained during three seasons of National Rugby League (NRL) matches to determine the need for further medical evaluation. If a concussion was suspected, players were removed from play and underwent a Head Injury Assessment, including the Sports Concussion Assessment Tool, Fifth Edition (SCAT5). Video footage was later examined to identify signs of possible LOC (i.e., observed LOC, no protective action in the fall, and unresponsiveness or lying motionless). Possible LOC was identified in 99 of the 1706 head impact events (5.8%). The median duration of apparent unresponsiveness was 4.2 sec (M = 7.4, standard deviation [SD] = 12.8, interquartile range [IQR] = 2.5–6.6). In the 661 athletes for whom SCAT5 data were available, those with possible LOC endorsed more SCAT5 symptoms and performed worse on Maddocks questions, Standardized Assessment of Concussion (SAC) total scores, orientation, immediate recall, concentration, and delayed recall. For the 255 players with medically diagnosed concussions, SCAT5 data were available for 245. Concussed players with possible LOC performed significantly worse on Maddocks questions. However, there were no group differences in SCAT5 symptom endorsement, SAC total scores, orientation, immediate recall, concentration, delayed recall, or the modified version of the Balance Error Scoring System (mBESS) total errors. Further, the presence or absence of possible LOC was not associated with number of games missed or time to medical clearance for match play. The duration of possible LOC was not associated with the number of games missed or time to medical clearance for match play. According to video review in NRL players, brief LOC might be more common than previously thought. The present study reveals possible LOC is not predictive of missed games or time to recover following concussion.

Keywords: brain injury; concussion; loss of consciousness; rugby; return to play; video analysis

[1]Department of Physical Medicine and Rehabilitation, Harvard Medical School, Boston, Massachusetts, USA.
[2]MassGeneral Hospital for Children Sports Concussion Program, Boston, Massachusetts, USA.
[3]Department of Physical Medicine and Rehabilitation, Spaulding Rehabilitation Hospital, Charlestown, Massachusetts, USA.
[4]Department of Physical Medicine and Rehabilitation, Schoen Adams Research Institute at Spaulding Rehabilitation, Charlestown, Massachusetts, USA.
[5]Sydney School of Health Sciences, Faculty of Medicine and Health, The University of Sydney, Camperdown, New South Wales, Australia.

*Address correspondence to: *Andrew J. Gardner, PhD, Sydney School of Health Sciences, Faculty of Medicine and Health, Susan Wakil Health Building, Western Avenue, The University of Sydney, Camperdown, New South Wales, 2050, Australia E-mail:* Andrew.Gardner@sydney.edu.au

Downloaded by Massachusetts Institute of Technology from www.liebertpub.com at 11/21/23. For personal use only.

Introduction

Most sport-related concussions occur without traumatic loss of consciousness (LOC).[1–3] When present, it is a striking and emotion-evoking visual sign of mild traumatic brain injury (mTBI). For several reasons, it is woven into the fabric of clinical lore that concussions with LOC represent more severe and longer-lasting injuries.[4] Historically, some assumed that LOC was necessary to define the injury[5,6] and described it as a characteristic feature of the injury.[7] In 1977, Parkinson conceptualized concussion more broadly than an injury involving LOC.[8] In addition, concussion grading scales, with their origins in contemporary medicine from the 1980s and 1990s, identified injuries with LOC as being the most severe form of concussion, necessitating different and more conservative medical management.[9–12] Further, the broader literature on TBI indicates that the duration of LOC is an indicator of the severity of TBI,[13,14] and that longer duration is associated with worse clinical outcome.[12,15] Finally, there have been a number of published studies with amateur[16–20] and professional[21,22] athletes illustrating that those who experienced LOC had a worse clinical outcome, as defined by a greater time before returning to sports.

There are reasons to question whether brief LOC, in the context of sport-related concussion, signifies a more severe injury to the brain or confers risk for worse clinical outcome. First, modern consensus and agreement statements do not recommend that concussions with brief LOC be conceptualized differently, or medically managed differently, than concussions without LOC.[3,23–25] Second, the duration of LOC in sports is typically measured in seconds, sometimes a few minutes, and thus the broader literature relating to prolonged LOC and coma, measured in hours or days, does not apply. Third, there are some mTBI studies of patients presenting to the emergency department for non-sport-related injuries indicating that those with LOC do not have worse acute or subacute neuropsychological outcome.[26,27] Finally, there are a number of published studies with athletes illustrating that those with LOC do not experience greater deficits on neuropsychological testing,[4,28,29] longer duration of clinical symptoms,[30–37] or slower return to sports.[31,38]

The literature on this topic is mixed and inconclusive, and there are only a small number of studies involving professional athletes.[21,22,29] An enormous methodological limitation of many prior studies is that LOC was self-reported by the athlete. In recent years, there has been considerable interest in identifying concussions, for research purposes, based on video review of the broadcast game footage. Video review studies of concussion have been conducted in diverse sports, such as American football,[39–43] Australian football,[44–47] ice hockey,[48–50] rugby union,[51,52] rugby league,[53–58] lacrosse,[59,60] and taekwondo.[61] Moreover, some professional sporting leagues now employ real-time video review during the game to try to identify possible injuries so the player can be promptly removed and medically evaluated.[44,62]

Leveraging these advancements in technology and league procedures, the present study was designed to determine whether elite-level rugby league players with on-field video evidence of possible LOC experience worse clinical outcome than concussed players who do not experience LOC. A recent study examined video-identified on-field motor incoordination in elite rugby league players and found that it was associated with subjectively experienced dizziness and balance problems, but not time to return to play.[58] For the present study, it was hypothesized that those with possible LOC would perform more poorly on orientation questions, cognitive testing, and balance testing on the sideline, immediately following injury, but there would be no significant difference in the number of days to return to play between those with versus without LOC.

Methods

Participants

The study included all athletes who participated in any match throughout the 2017–2019 seasons of the National Rugby League (NRL). Each season consisted of 201 matches (192 regular season and 9 post-season). Head impact events were identified by either the video sideline injury surveillance system or team doctor. Head impact events were recorded and uploaded to the *Gameplan* application. The athletes consented *a priori*, as part of their collective bargaining agreement, to allow for use of their deidentified injury data in research endorsed by the Rugby League Research Committee. This study was approved by the Institutional Human Ethics Committee at the University of Newcastle (Ref No. H-2012-0344). These data have been analyzed in prior research examining the association between concussion and on-field motor incoordination.[58]

NRL personnel examine video footage in real time in the stadium and identify and record blows to the head or body that might cause injury (termed a ''head impact event'') for every match. Medical personnel can prompt sideline operators to review incidents via replay. Sideline operators are charged with identifying and tagging all head impact events and match play injuries within the video surveillance system. Consequently, head impact events included in the database were identified by either medical personnel, sideline operators, or both. Post-match video clips included footage of the head impact events captured from multiple angles and in normal speed and slow motion and typically ranged from 1 to 5 min. The senior author (AJG) reviewed all clips in high definition via an Apple iPhone 10S (5.85" diagonally) and coded signs associated with concussion.[53]

Downloaded by Massachusetts Institute of Technology from www.liebertpub.com at 11/21/23. For personal use only.

Head injury assessments

In the NRL, an off-field medical assessment process (i.e., a Head Injury Assessment) of a player is triggered when a head impact event occurs that results in video signs of potential concussion, or there is expressed concern from the team medical personnel. For example, if a player shows evidence of possible LOC, no protective action during a fall to the ground, motor incoordination upon standing, a blank or vacant stare, or atypical behavior, he is immediately removed from the match for a medical assessment. As part of the Head Injury Assessment process (a 15-min off-field assessment), the Sport Concussion Assessment Tool, Fifth Edition (SCAT5)[63] is administered by the team physician using a tablet-based web-hosted CSx data platform.[62,64] The team physician uses the results from history taking, observation, clinical examination, and the SCAT5 to render a diagnosis of concussion or suspected concussion. Any player presumed to have a concussion is not permitted to return to play in that game.

Defining unresponsiveness/Apparent loss of consciousness

We applied the following criteria to determine possible LOC. To meet criterion 1, a player had to have been removed from a match by medical personnel (e.g., team physician) due to suspected LOC following a head impact event and be rated as showing at least one of two possible signs of LOC on post-match video review (i.e., no protective action in the fall or unresponsiveness or lying motionless). To meet criterion 2, a player had to show both possible signs of LOC on post-match video review (i.e., no protective action in the fall and unresponsiveness or lying motionless) irrespective of whether the player was removed from a match by medical personnel due to suspected LOC following a head impact. Head impacts meeting either criterion 1 or criterion 2 were classified as possible LOC.

Defining duration of unresponsiveness

During the post-match review of all video clips, the senior author (AJG) timed the duration of unresponsiveness. The duration of unresponsiveness was calculated using a stopwatch. The stopwatch was started from the moment of impact and stopped when the athlete showed video evidence of an initial voluntary movement (i.e., moved an upper or lower limb or appeared to communicate). The time was recorded in seconds, to two decimal places (i.e., hundredths of a second).

Measures

The SCAT5[63] is a standardized acute clinical assessment for athletes suspected of concussion and has been described by the International Concussion in Sport Group.[63,65] The SCAT, in multiple iterations, is widely used in professional and amateur sports.[66–70] The SCAT5 includes an immediate (or on-field) assessment and an office (or off-field assessment). The immediate assessment includes ''red flags,'' observable signs (e.g., lying motionless on the playing field), sport-specific orientation and amnesia assessed using the five Maddocks questions, the Glasgow Coma Scale, and a cervical spine assessment. The office (off-field) assessment includes: 1) background information, such as demographic variables, self-reported concussion history, potential outcome modifiers, and medications; 2) a concussion symptom evaluation consisting of 22 symptoms that are graded on a dimensional scale from 0 (none) to 6 (severe); 3) a cognitive assessment based on the Standardized Assessment of Concussion (SAC),[71] which assesses time orientation, immediate memory, concentration, and delayed recall components; 4) a neurological screening evaluation involving a cervical/neck examination, eye movements, finger-to-nose test, and tandem gait; and 5) a measure of postural stability using the modified version of the Balance Error Scoring System (mBESS).[72]

The symptom checklist was modified from its original format for the Head Injury Assessment. The symptom ''trouble falling asleep (if applicable)'' was removed from the symptom checklist for use in Head Injury Assessments and replaced with the symptom ''excessive tiredness.'' This modification was initiated by World Rugby on the basis that the symptom of trouble falling asleep was not relevant to the acute sideline assessment, but that excessive tiredness was a relevant symptom. This change was adopted and incorporated into the CSx App for all sports using this platform, including the NRL. The test components included in this app and available for this study included the orientation and amnesia testing (Maddocks questions), symptom scale, cognitive testing (SAC), and mBESS.

Statistical analysis

Analyses examined between-group differences among those with and without possible LOC. Chi-square tests assessed differences between categorical variables and odds ratios (ORs) were computed to characterize effect sizes.[73,74] Mann-Whitney U tests were used for non-normally distributed variables (e.g., days to be medically cleared, SCAT5 variables) and Cohen's d effects sizes were computed using test statistics.[73–75] Effect sizes were interpreted according to conventional guidelines, i.e., $d=0.2$ (small), $r=0.5$ (medium), and $r=0.8$ (large).[74] SCAT5 analyses excluded SAC Immediate Memory, Delayed Memory, and Total Score data from a very small number of players who were administered the 10-word list because nearly all of the sample

Downloaded by Massachusetts Institute of Technology from www.liebertpub.com at 11/21/23. For personal use only.

completed the 5-word list. Exploratory analyses examining associations between duration of unresponsiveness among those with possible LOC and games missed, days until medical clearance, and SCAT5 variables were also undertaken. To adjust for multiple comparisons, the Benjamini-Hochberg procedure was performed and a false discovery rate of 10% was applied resulting in an alpha level of 0.01.[76] Analyses were performed in IBM SPSS (IBM Corp.; released 2021, IBM SPSS Statistics V.28).

Results

The database included 1706 head impact events, of which 727 (42.6%) resulted in a Head Injury Assessment and 255 (14.9%) in medically diagnosed concussions. There were 99 players who had evidence of possible LOC as determined by our two conditional criteria (5.8% of all head impact events). Forty-two of the possible LOC head impact events met criterion 1 (i.e., were identified during the matches by the medical staff for suspected LOC and confirmed during post-match video review; 42.4%). Ninety-two of the possible LOC head impact events met criterion 2 (i.e., showed both no protective action and no responsiveness/motionless signs on video review; 92.9%). Frequencies of observed criteria among head impact events are shown in Table 1. The distribution of possible LOC cases is shown in Figure 1. Among head impact events that met possible LOC criteria, including those with signs identified during post-match review ($n = 99$, 5.8%), 93 (93.9%) underwent a Head Injury Assessment and 68 (68.7%) were medically diagnosed with concussion. For head impact events that warranted removal from play due to suspected LOC documented by medical personnel ($n = 53/1706$, 3.1%), 100% of the cases underwent Head Injury Assessments and 51 (96.2%) were medically diagnosed with concussions. Of those who underwent Head Injury Assessments ($n = 727$), those with possible LOC were more likely to be diagnosed with concussion than those without possible LOC (χ^2 [1, 727] = 81.86, $p < 0.001$, OR = 7.67; 95% confidence interval [CI] = 4.69-12.54).

Possible loss of consciousness and SCAT5 scores

Total sample undergoing sideline evaluations. SCAT5 data were available for 661 athletes who underwent sideline evaluations. SCAT5 component scores by group (presence/absence of possible LOC) are displayed in Table 2. Participants with possible LOC endorsed more SCAT5 symptoms and performed worse on Maddocks questions, SAC total scores, orientation, immediate recall, concentration, and delayed recall. There were no group differences in mBESS errors. As can be seen in Table 3, those with possible LOC endorsed the majority of SCAT5 symptoms at higher rates than those without possible LOC (17/22 items, 77.3%). Notably, ''sensitivity to noise,'' ''don't feel right,'' ''difficulty concentrating,'' ''difficulty remembering,'' ''excessive tiredness,'' and ''feeling more emotional'' showed ORs greater than 4. The frequencies of incorrect responses on Maddocks questions for those with and without possible LOC are displayed in Table 4.

Medically diagnosed concussions. For the 255 medically diagnosed concussions, 245 had sideline evaluation SCAT5 data available. SCAT5 component scores by group (presence/absence of possible LOC) are displayed in Table 5. Participants with possible LOC performed significantly worse on Maddocks questions. There were no group differences in SCAT5 symptom endorsement, SAC total scores, orientation, immediate memory, concentration, delayed recall, and mBESS total errors. There were no group differences in individual symptom endorsement rates (Table 3). Those with possible LOC made more errors on individual Maddocks questions at higher rates than those without possible LOC (Table 4).

Table 1. Indicators of Loss of Consciousness

		Video signs				
	Physician identified LOC	*No protective action*	*Loss of responsiveness/ Motionless*	*Criterion 1*	*Criterion 2*	*Possible LOC*
All head impact events ($n = 1706$)	53 (3.1%)[a]	109 (6.4%)	136 (8.0%)	42 (2.5%)	92 (5.4%)	**99 (5.8%)**
Head Injury Assessments ($n = 727$)	53 (7.3%)[b]	109 (15.0%)	136 (18.7%)	42 (5.8%)	92 (12.7%)	**99 (13.6%)**
Medically diagnosed concussions ($n = 255$)	51 (20.0%)[c]	67 (26.3%)	81 (31.8%)	41 (16.1%)	61 (23.9%)	**68 (26.7%)**

[a]Physician after-match reports were completed for 696/1706 head impact events documented in the database.
[b]Of the 727 Head Injury Assessments, 66 did not have the accompanying physician report in the database.
[c]Of the 255 medically diagnosed concussions, 10 did not include a physician report.
Criterion 1: physician identified LOC and one or more video signs; Criterion 2: appearance of both video signs.
LOC, loss of consciousness.

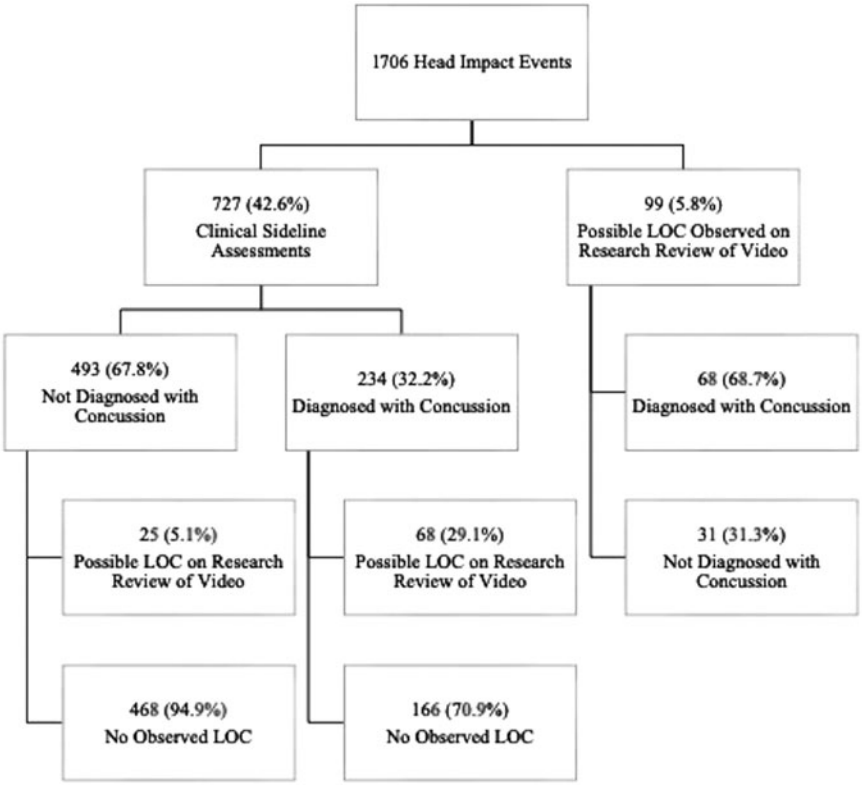

FIG. 1. The clinical sideline assessments are called Head Injury Assessments in the National Rugby League. There were 255 athletes with medically diagnosed concussions included in the database. There were 21 athletes who were diagnosed with concussion who did not have clinical sideline assessment results in the database, thus 234 of those with diagnosed concussions had clinical assessment data available. LOC, loss of consciousness.

Of the Head Injury Assessments that did not result in a medically diagnosed concussion, 451 had sideline evaluation SCAT5 data available. SCAT5 component scores by group (presence/absence of possible LOC) are displayed in Table 6. There were no significant differences between groups in symptom endorsement, Maddocks questions, SAC total scores, orientation, immediate memory, concentration, delayed recall, and mBESS total errors. Similarly, they did not differ in frequencies of incorrect responses on Maddocks questions (Table 4).

Table 2. SCAT5 Component Scores by Presence Versus Absence of Possible Loss of Consciousness for the Full Sample of Subjects Who Underwent Sideline Medical Evaluation

	Possible LOC					*No possible LOC*							
Test component	n	*Mean*	*SD*	*Mdn*	*IQR*	n	*Mean*	*SD*	*Mdn*	*IQR*	*U*	p	d
Symptom severity	90	10.2	16.9	2.0	0-13.3	571	3.8	9.6	1.0	0-3	31987.0	**<0.01**	0.29
Number of symptoms	90	4.3	5.7	1.5	0-7	571	1.9	3.6	1.0	0-2	32002.0	**<0.01**	0.30
Maddocks questions total	90	4.2	1.5	5.0	4-5	571	4.8	0.8	5.0	5-5	20271.0	**<0.01**	0.25
SAC total score	89	24.3	7.0	27.0	23-29	554	27.0	4.0	28.0	26-29	19307.0	**<0.01**	0.26
Orientation	90	4.0	1.5	5.0	4-5	571	4.6	0.8	5.0	4-5	19473.5	**<0.01**	0.29
Immediate Memory	90	13.2	3.3	15.0	13-15	571	14.2	2.1	15.0	14-15	21700.0	**<0.01**	0.19
Concentration	90	3.5	1.5	4.0	3-5	571	4.0	1.1	4.0	3-5	22117.5	0.03	0.17
Delayed Recall	88	3.5	1.9	4.0	2-5	552	4.2	1.3	5.0	4-5	20063.5	**<0.01**	0.21
mBESS total errors	90	4.1	3.6	3.0	1-6.3	571	3.7	3.0	3.0	1-5	26360.5	0.69	0.10

SAC total score available for 643 athletes.

IQR, interquartile range; LOC, loss of consciousness; mBESS, Modified Balance Error Scoring System; Mdn, median; SAC, Standardized Assessment of Concussion; SD, standard deviation.

Downloaded by Massachusetts Institute of Technology from www.liebertpub.com at 11/21/23. For personal use only.

Table 3. Symptom Endorsement by Presence Versus Absence of Possible Loss of Consciousness

	Full sample						*Medically diagnosed concussions*					
	Possible LOC						*Possible LOC*					
Symptom	*Yes (n = 90)*	*No (n = 571)*	χ^2	p	*OR*	*95% CI*	*Yes (n = 68)*	*No (n = 177)*	χ^2	p	*OR*	*95% CI*
Headache	35 (38.9)	139 (24.3)	8.5	**<0.01**	2.0	1.2-3.2	32 (47.1)	91 (51.4)	0.37	0.54	0.84	0.5-1.5
"Pressure in head"	29 (32.2)	87 (15.2)	15.5	**<0.01**	2.7	1.6-4.4	25 (36.8)	70 (39.5)	0.16	0.69	0.89	0.5-1.6
Neck pain	21 (23.3)	96 (16.8)	2.3	0.13	1.5	0.9-2.6	19 (27.9)	43 (24.3)	0.35	0.56	1.21	0.6-2.3
Nausea or vomiting	11 (12.2)	29 (5.1)	7.0	**<0.01**	2.6	1.3-5.4	11 (16.2)	24 (13.6)	0.28	0.60	1.23	0.6-2.7
Dizziness	27 (30.0)	102 (17.9)	7.3	**<0.01**	2.0	1.2-3.2	25 (36.8)	81 (45.8)	1.62	0.20	0.70	0.4-1.2
Blurred vision	11 (12.2)	72 (12.6)	0.0	0.92	1.0	0.5-1.9	10 (14.7)	46 (26.0)	3.55	0.06	0.49	0.2-1.0
Balance problems	19 (21.1)	43 (7.5)	16.9	**<0.01**	3.3	1.8-6.0	17 (25.0)	36 (20.3)	0.63	0.43	1.31	0.7-2.5
Sensitivity to light	6 (6.7)	19 (3.3)	2.4	0.12	2.1	0.8-5.3	6 (8.8)	15 (8.5)	0.01	0.93	1.05	0.4-2.8
Sensitivity to noise	6 (6.7)	8 (1.4)	10.4	**<0.01**	5.0	1.7-14.9	6 (8.8)	6 (3.4)	3.11	0.08	2.76	0.9-8.9
Feeling slowed down	22 (24.4)	59 (10.3)	14.4	**<0.01**	2.8	1.6-4.9	21 (30.9)	51 (28.8)	0.10	0.75	1.1	0.6-2.0
Feeling like "in a fog"	21 (23.3)	50 (8.8)	17.2	**<0.01**	3.1	1.8-5.6	19 (27.9)	51 (28.8)	0.02	0.89	1.0	0.5-1.8
"Don't feel right"	35 (38.9)	70 (12.3)	41.3	**<0.01**	4.6	2.8-7.5	33 (48.5)	66 (37.3)	2.58	0.11	1.6	0.9-2.8
Difficulty concentrating	26 (28.9)	50 (8.8)	31.0	**<0.01**	4.2	2.5-7.3	24 (35.3)	47 (26.6)	1.82	0.18	1.5	0.8-2.8
Difficulty remembering	26 (28.9)	46 (8.1)	34.8	**<0.01**	4.6	2.7-8.0	24 (35.4)	47 (26.6)	1.82	0.18	1.5	0.8-2.8
Fatigue or low energy	19 (21.1)	63 (11.0)	7.3	**<0.01**	2.3	1.3-3.8	17 (25.0)	39 (22.0)	0.25	0.62	1.2	0.6-2.3
Confusion	20 (23.6)	38 (6.7)	23.5	**<0.01**	4.0	2.2-7.3	18 (26.5)	37 (20.9)	0.87	0.35	1.4	0.7-2.6
Drowsiness	14 (15.6)	32 (5.6)	11.9	**<0.01**	3.1	1.6-6.1	14 (20.6)	28 (15.8)	0.79	0.38	1.4	0.7-2.8
Excessive tiredness[a]	7 (7.8)	11 (1.9)	10.1	**<0.01**	4.3	1.6-11.4	7 (10.3)	8 (4.5)	2.85	0.09	2.4	0.8-7.0
More emotional	8 (8.9)	13 (2.3)	11.1	**<0.01**	4.2	1.7-10.4	8 (11.8)	12 (6.8)	1.63	0.20	1.8	0.7-4.7
Irritability	12 (13.3)	24 (4.2)	12.6	**<0.01**	3.5	1.7-7.3	10 (14.7)	17 (9.6)	1.30	0.25	1.6	0.7-3.7
Sadness	4 (4.4)	14 (2.5)	1.2	0.28	1.9	0.6-5.8	4 (5.9)	14 (7.9)	0.30	0.59	0.7	0.2-2.3
Nervous or anxious	7 (7.8)	21 (3.7)	3.2	0.07	2.2	0.9-5.4	7 (10.3)	16 (9.0)	0.09	0.76	1.2	0.5-2.9

[a]The symptom "excessive tiredness" is not part of the official version of the SCAT5 (Sports Concussion Assessment Tool, Fifth Edition). It was added to the assessment platform based on a recommendation by World Rugby to replace the symptom "trouble falling asleep," which is not a relevant symptom for day of injury medical evaluations. This symptom was used in the calculation of the two symptom total scores.

The numbers represent the frequency and the percentages (in parentheses) of the sample endorsing each symptom.

CI, confidence interval; LOC, loss of consciousness; OR, odds ratio.

Time to medical clearance to return to sports

For the 255 medically diagnosed concussions, 229 had data available for missed games (89.8%). Due to non-normality in the number of games missed variable, the variable was transformed to a binary state (0 vs. 1+ games missed). For those with possible LOC, 21.5% missed one or more games and for those with no observed LOC, 14.0% missed one or more games (χ^2 [1, 229] = 1.94, $p = 0.164$, OR = 1.68, 95% CI = 0.81-3.52).

Of the 255 medically diagnosed concussions, 230 had data available for medical clearance for match play (in days; 90.2%). The Mann-Whitney U test examining days to medical clearance by presence of possible LOC ($n = 65$, mean [M] = 7.8 days, SD = 5.8, median

Table 4. On-Field Disorientation and Amnesia as Measured by the Maddocks Questions of the SCAT5

	Total sample			*No diagnosed concussion*			*Medically diagnosed concussion*		
		Possible LOC			*Possible LOC*			*Possible LOC*	
Maddocks questions	*All athletes (n = 661)*	*No (n = 571)*	*Yes (n = 90)*	*All athletes (n = 451)*	*No (n = 429)*	*Yes (n = 22)*	*All athletes (n = 245)*	*No (n = 177)*	*Yes (n = 68)*
Which half is it now?	19 (2.9)	11 (1.9)	8 (8.9)	8 (1.8)	8 (1.9)	0 (0.0)	12 (4.9)	4 (2.3)	8 (11.8)
Did your team win the last game?	35 (5.3)	22 (3.9)	13 (14.4)	8 (1.8)	8 (1.9)	0 (0.0)	29 (11.8)	16 (9.0)	13 (19.1)
Who scored last in this match?	38 (5.7)	23 (4.0)	15 (16.7)	11 (2.4)	10 (2.3)	1 (4.5)	29 (11.8)	15 (8.5)	14 (20.6)
What team did you play last week?	55 (8.3)	33 (5.8)	22 (24.4)	9 (2.0)	9 (2.1)	0 (0.0)	49 (20.0)	27 (15.3)	22 (32.4)
What venue are we at today?	20 (3.0)	10 (1.8)	10 (11.1)	9 (2.0)	8 (1.9)	1 (4.5)	12 (4.9)	3 (1.7)	9 (13.2)
Missed one or more questions	64 (9.7)	39 (6.8)	25 (27.8)	12 (2.7)	11 (2.6)	1 (4.5)	56 (22.9)	32 (18.1)	24 (35.3)

The numbers in the table represent the frequency (percentage) of athletes answering the question incorrectly. For those with medically diagnosed concussions, 10 cases were missing data.

LOC, loss of consciousness; SCAT5, Sports Concussion Assessment Tool, Fifth Edition.

Downloaded by Massachusetts Institute of Technology from www.liebertpub.com at 11/21/23. For personal use only.

Table 5. SCAT5 Component Scores by Presence Versus Absence of Possible Loss of Consciousness Among Those With a Medically Diagnosed Concussion

	Possible LOC (n = 63)				*No LOC (n = 182)*						
Test component	*Mean*	*SD*	*Mdn*	*IQR*	*Mean*	*SD*	*Mdn*	*IQR*	U	p	d
Symptom severity	12.7	18.9	4.0	0-18	9.4	11.8	5.5	1-13	5671.5	0.90	0.02
Number of symptoms	5.3	6.2	3.0	0-8	4.8	4.9	3.0	1-7	5556.0	0.71	0.05
Maddocks questions total	4.0	1.6	5.0	3-5	4.6	1.0	5.0	5-5	4700.0	<0.01	0.28
SAC total score[a]	23.1	7.8	25.0	22-29	25.0	4.3	26.0	23-28	5251.0	0.57	0.13
Orientation	3.8	1.6	4.0	3-5	4.3	1.2	5.0	4-5	4894.5	0.06	0.22
Immediate Memory	12.8	3.8	15.0	12-15	13.8	2.4	15.0	14-15	5090.5	0.14	0.17
Concentration	3.4	1.7	4.0	2-5	3.5	1.3	4.0	3-5	5809.5	0.87	0.02
Delayed Recall[b]	3.1	2.0	4.0	2-5	3.5	1.6	4.0	3-5	5070.5	0.43	0.10
mBESS total errors	4.2	3.6	3.0	1-7	5.2	3.9	4.5	2-7	4819.0	0.06	0.24

[a]For Possible LOC, $n = 62$; for No LOC, $n = 178$.
[b]For Possible LOC, $n = 61$; for No LOC, $n = 178$.
SCAT5 data were not available for 5 players.
IQR, interquartile range; LOC, loss of consciousness; mBESS, modified Balance Error Scoring System; SAC, Standardized Assessment of Concussion; SCAT5, Sports Concussion Assessment Tool, Fifth Edition; SD, standard deviation.

Downloaded by Massachusetts Institute of Technology from www.liebertpub.com at 11/21/23. For personal use only.

[Mdn] = 6.0, interquartile range [IQR] = 5–10, range = 0–40, 90% cumulative return = 13 days), and no observed LOC ($n = 165$, M = 7.4 days, SD = 7.7, Mdn = 6.0, IQR = 5–9, range 0–79, 90% cumulative return = 12 days) was not significant ($U = 5677.0$, $p = 0.486$, $d = 0.09$).

Duration of unresponsiveness

For the 99 head impact events with possible LOC, duration of unresponsiveness was available for 97 cases (97.9%). The mean duration of apparent unresponsiveness was 7.4 sec (Mdn = 4.2, SD = 12.8, IQR = 2.5–6.6, Range 0.4–93.4). Those with medically diagnosed concussions showed a longer duration of unresponsiveness ($n = 66$, M = 9.1 sec, Mdn = 4.9, SD = 15.2, IQR = 2.8–7.9) than those without medically diagnosed with concussions ($n = 31$, M = 4.0 sec, Mdn = 3.8, SD = 2.9, IQR = 1.9–4.4; $U = 1340.5$, $p = 0.014$, $r = 0.25$).

Missed games and duration of unresponsiveness were available for 63 medically diagnosed concussions. Medically diagnosed concussions that resulted in one or more missed games did not show a longer duration of unresponsiveness ($n = 14$, Mdn = 5.1, SD = 29.1, IQR = 2.9–13.5) than those that did not miss a game ($n = 49$, Mdn = 4.9, SD = 7.6, IQR = 2.8–7.9; $U = 375.5$, $p = 0.591$, $r = 0.07$). Duration of LOC was not associated with medical clearance for match play ($r = 0.064$, $p = 0.620$, 95% CI = −0.19 to 0.31).

Discussion

LOC is a dramatic on-field sign of sport-related concussion. Historically, the presence of LOC was conceptualized as a more severe injury that required different and more conservative medical management.[9–12] In the modern era, concussions with LOC are not expected to be managed differently than concussions without LOC.[3, 23–25] In the present study, players were removed from the match following a blow to the head 727 times over the course of three seasons. Of those, 99 of the 727 Head Injury Assessments (13.6%) had video evidence of possible LOC. Of the total number of those who underwent a

Table 6. SCAT5 Component Scores by Presence Versus Absence of Possible Loss of Consciousness Among Those Who Were Not Medically Diagnosed With Concussion

	Possible LOC (n = 20)				*No LOC (n = 431)*						
Test component	*Mean*	*SD*	*Mdn*	*IQR*	*Mean*	*SD*	*Mdn*	*IQR*	U	p	d
Symptom severity	3.9	8.7	0.0	0-3	1.7	7.3	0.0	0-1	4393.0	0.87	0.01
Number of symptoms	1.5	2.6	0.0	0-3	0.9	2.0	0.0	0-1	4429.5	0.81	0.02
Maddocks questions total	4.9	0.5	5.0	5-5	4.9	0.7	5.0	5-5	4207.0	0.52	0.02
SAC total score[a]	27.4	3.0	28.0	27-29	27.8	3.5	29.0	27-30	3637.0	0.32	0.09
Orientation	4.8	0.4	5.0	4-5	4.7	0.7	5.0	5-5	4140.0	0.67	0.03
Immediate Memory	14.4	1.3	15.0	14-15	14.4	1.9	15.0	14-15	3650.0	0.37	0.07
Concentration	3.9	1.1	4.0	3-5	4.2	1.0	4.0	4-5	3918.5	0.21	0.11
Delayed Recall[b]	4.4	1.1	5.0	4-5	4.4	1.0	5.0	4-5	4185.0	0.96	0.00
mBESS total errors	3.9	4.1	3.0	1-5	3.1	2.3	3.0	1-5	4533.0	0.69	0.04

[a]For Possible LOC, $n = 20$; for No LOC, $n = 418$.
[b]For Possible LOC, $n = 20$; for No LOC, $n = 416$.
SCAT5 data were not available for 13 players.
IQR, interquartile range; LOC, loss of consciousness; mBESS, modified Balance Error Scoring System; SAC, Standardized Assessment of Concussion; SCAT5, Sports Concussion Assessment Tool, Fifth Edition; SD, standard deviation.

Downloaded by Massachusetts Institute of Technology from www.liebertpub.com at 11/21/23. For personal use only.

medical evaluation on the sideline, 255 of the 727 (35.1%) players undergoing Head Injury Assessments were ultimately diagnosed as having a concussion. Some prior studies with high school,[1,77] college,[1,2] and professional athletes[78] suggest that approximately 10% or fewer of concussions have LOC as a sign of injury—but those studies relied on LOC being self-reported by the athlete. In contrast, some large-scale published video review studies have identified possible or probable LOC in approximately 20–75% of concussions.[40,54,79,80]

In the present study, across three NRL seasons, there were 255 medically diagnosed concussions. Of those, 68 (26.7%) had visible evidence of possible LOC on video. The results from the present study, and other video review studies, suggest that brief LOC might be more common than previously thought. The estimated duration of possible LOC in most cases in this study was only a few seconds. The association between the duration of possible LOC and clinical outcome has not been well characterized in sport-related concussion. Methodologically, the majority of research has examined LOC as a binary variable (i.e., present or not present), as opposed to duration, in relation to clinical outcomes.[81] Duration of LOC is not frequently reported.[82] Results from the present study did not reveal an association between the duration of possible LOC and two clinical outcomes of interest, games missed and medical clearance for return to match play.

In the current study, the presence of possible LOC was associated with worse acute mental status, as determined by missing one or more questions on the SCAT5 Maddocks questions. Maddocks questions have not shown a high sensitivity for concussion in prior research involving rugby athletes.[65] In a recent study of these same rugby league players, focused on whether on-field motor incoordination was associated with worse outcome, athletes observed to show motor incoordination performed worse on the Maddocks questions than those without motor incoordination. However, this finding did not hold among a subset of the sample with medically diagnosed concussion.[58] In the present study, the concussed athletes with possible LOC had greater difficulty providing accurate responses to the majority of Maddocks questions than those without possible LOC.

Limitations

This study has several methodological limitations. First and foremost, there could be a degree of both circularity and confirmatory bias that enhanced the association between video evidence of LOC and a medical diagnosis of concussion. Specifically, if the medical staff observes a blow to the head, followed by a player appearing to display LOC, that sets up a higher expectation for concussion. This could influence how the physician interprets the examination findings, possibly lowering the threshold for diagnosing a concussion. Moreover, the SCAT5 test results are not independent of the medical diagnosis of concussion. That is, the SCAT5 is used, in part, to make the diagnosis. In addition, the NRL match day concussion policy mandates immediate and permanent removal of an athlete who displays evidence of ''loss of consciousness,'' ''suspected loss of consciousness,'' or ''no protective action.'' The process of permanently removing an athlete from play may also lower the threshold for diagnosing a concussion. Second, definitively determining the presence of LOC is usually not possible using video review. Under most circumstances, LOC can only be determined definitively by a health care provider, such as an athletic trainer or physician, who is physically present and observes the unconsciousness.

Third, calculating the duration of unresponsiveness using video review has methodological limitations related to the use of a handheld stopwatch, which requires judgment relating to the start and stop times (i.e., presumed onset of LOC or unresponsiveness to resolution) and the reliability of this procedure is uncertain. Relatedly, the use of an Apple iPhone (screen size: 5.85" diagonally) rather than a desktop monitor may have conceivably reduced the rater's ability to detect possible signs of LOC. Fourth, all the videos in this study were viewed and coded by a single rater (AJG), consistent with some of our prior studies.[56–58,83] The senior author and rater has extensive experience with concussion, video analysis, and rugby given his sideline assistance to medical staff during rugby league matches and his brain injury research program. The results could differ, somewhat, if another person coded the videos for consensus purposes. Fifth, there could be some disparities in how medical staff from each rugby team identify and diagnose concussion. Video analysis was conducted without in-game time pressure, thus our findings cannot be assumed to generalize to real-time video identification and analysis. Lastly, it cannot be assumed that the findings from this study would generalize to amateur contact or collision sports, and certainly not to other mechanisms of mild traumatic brain injury outside of sports.

Conclusions

Over the course of three NRL seasons, possible LOC was observed in approximately 1 in 20 (5.8%) blows to the head identified on video, 1 in 10 (13.6%) of those medically evaluated on the sideline for concussion, and 1 in 4 (26.7%) of those medically diagnosed with concussion. Of those medically evaluated, having possible LOC was clearly associated with worse acute clinical outcome, as measured by orientation and amnesia on the sideline, symptom reporting, and cognitive testing on the SCAT5. However, for those ultimately medically diagnosed with

a concussion, possible LOC was not clearly associated with worse clinical outcomes. Those with possible LOC were more likely to show disorientation and amnesia on the sideline, but they did not report significantly more symptoms, perform worse on cognitive testing, perform worse on balance testing, or take longer to clinically recover and return to play. These results support current practice guidelines to treat concussions based on clinical evaluation results and individual needs, and to not treat those who experience LOC differently than those who do not experience LOC.

Authors' Contributions

GI conceptualized the study, assisted with the literature review, conceptualized the statistical analyses, helped conduct the statistical analyses, drafted sections of the manuscript, edited the manuscript, and approved the final manuscript. CEG assisted with reviewing the literature, conducted the statistical analyses, drafted portions of the manuscript, edited the manuscript, and approved the final manuscript. AKK assisted with reviewing the literature, drafted portions of the manuscript, edited the manuscript, and approved the final manuscript. AJG served as the principal investigator, secured regulatory ethics for the study, helped conceptualize the study, examined and coded all of the videos used in the study, created the database, edited the manuscript, and approved the final manuscript. All authors contributed to the manuscript and approved the submitted version.

Funding Information

There was no direct or sponsored funding for this study. The authors acknowledge unrestricted philanthropic support from the National Rugby League. GI acknowledges unrestricted philanthropic support from ImPACT Applications, Inc., the Mooney-Reed Charitable Foundation, and the Spaulding Research Institute. AG acknowledges unrestricted philanthropic support from the Tooth Foundation. The donors and funders were not involved in the study design, collection, analysis, interpretation of data, the writing of this article, or the decision to submit it for publication.

Author Disclosure Statement

GI has a clinical and consulting practice in forensic neuropsychology, including expert testimony, involving individuals who have sustained mild traumatic brain injuries (including athletes). He has received research funding from several test publishing companies, including ImPACT Applications, Inc., CNS Vital Signs, and Psychological Assessment Resources (PAR, Inc.). He has received research support from the Harvard Integrated Program to Protect and Improve the Health of NFLPA Members, and grant support from the National Football League. He serves as a scientific advisor for Sway Operations, LLC, Highmark, Inc., and NanoDX®.

AG serves as a scientific advisor for HitIQ, Ltd. He has a clinical practice in neuropsychology involving individuals who have sustained sport-related concussion (including current and former athletes). He has been a contracted concussion consultant to Rugby Australia since July 2016. He has received travel funding or been reimbursed by professional sporting bodies, and commercial organizations for discussing or presenting sport-related concussion research at meetings, scientific conferences, workshops, and symposiums. Previous grant funding includes the NSW Sporting Injuries Committee; the Brain Foundation (Australia); an Australian-American Fulbright Commission Postdoctoral Award; an NHMRC early research career fellowship; a Hunter New England Local Health District, Research, Innovation and Partnerships Health Research & Translation Centre and Clinical Research Fellowship Scheme; and the Hunter Medical Research Institute (HMRI), supported by Jennie Thomas, and the HMRI, supported by Anne Greaves.

References

1. Guskiewicz KM, Weaver NL, Padua DA, et al. Epidemiology of concussion in collegiate and high school football players. Am J Sports Med 2000;28(5):643–650.
2. McCrea M, Guskiewicz KM, Marshall SW, et al. Acute effects and recovery time following concussion in collegiate football players: the NCAA Concussion Study. JAMA 2003;290(19):2556–2563.
3. McCrory P, Meeuwisse W, Dvořák J, et al. Consensus statement on concussion in sport—the 5th International Conference on Concussion in Sport held in Berlin, October 2016. Br J Sport Med 2017;51(11):838–847.
4. Collins MW, Iverson GL, Lovell MR, et al. On-field predictors of neuropsychological and symptom deficit following sports-related concussion. Clin J Sport Med 2003;13(4):222–229.
5. Ward AA Jr. The physiology of concussion. Clin Neurosurg 1964;12:95–111.
6. Symonds C. Concussion and its sequelae. Lancet 1962;279:1–5.
7. Symonds CP. Observations on the differential diagnosis and treatment of cerebral states consequent upong head injuries. Br Med J 1928;2(3540):829–832.
8. Parkinson D. Concussion. Mayo Clin Proc 1977;52(8):492–496.
9. Cantu RC. Guidelines for return to contact sports after a cerebral concussion. Phys Sportsmed 1986;14(10):75–83.
10. Cantu RC. Cerebral concussion in sport. Management and prevention. Sports Med 1992;14(1):64–74.
11. Kelly JP, Nichols JS, Filley CM, et al. Concussion in sports. Guidelines for the prevention of catastrophic outcome. JAMA 1991;266(20):2867–2869.
12. Practice parameter: the management of concussion in sports (summary statement). Report of the Quality Standards Subcommittee. Neurology 1997;48(3):581–585.
13. Leclerc S, Lassonde M, Delaney JS, et al. Recommendations for grading of concussion in athletes. Sports Med 2001;31(8):629–636.
14. Kelly JP. Loss of consciousness: pathophysiology and implications in grading and safe return to play. J Athl Train 2001;36(3):249–252.
15. Whyte J, Cifu D, Dikmen S, et al. Prediction of functional outcomes after traumatic brain injury: a comparison of 2 measures of duration of unconsciousness. Arch Phys Med Rehabil 2001;82(10):1355–1359.
16. Chermann JF, Klouche S, Savigny A, et al. Return to rugby after brain concussion: a prospective study in 35 high level rugby players. Asian J Sports Med 2014;5(4):e24042.
17. Guskiewicz KM, McCrea M, Marshall SW, et al. Cumulative effects associated with recurrent concussion in collegiate football players: the NCAA Concussion Study. JAMA 2003;290(19):2549–2555.
18. Asplund CA, McKeag DB, Olsen CH. Sport-related concussion: factors associated with prolonged return to play. Clin J Sport Med 2004;14(6):339–343.

Downloaded by Massachusetts Institute of Technology from www.liebertpub.com at 11/21/23. For personal use only.

19. Miller JH, Gill C, Kuhn EN, et al. Predictors of delayed recovery following pediatric sports-related concussion: a case-control study. J Neurosurg Pediatr 2016;17(4):491–496.
20. Heyer GL, Schaffer CE, Rose SC, et al. specific factors influence postconcussion symptom duration among youth referred to a sports concussion clinic. J Pediatr 2016;174, 33.e32–38.e32.
21. Benson BW, Meeuwisse WH, Rizos J, et al. A prospective study of concussions among National Hockey League players during regular season games: the NHL-NHLPA Concussion Program. CMAJ 2011;183(8):905–911.
22. Pellman EJ, Viano DC, Casson IR, et al. Concussion in professional football: injuries involving 7 or more days out—Part 5. Neurosurgery 2004;55(5):1100–1119.
23. McCrory P, Meeuwisse W, Aubry M, et al. Consensus statement on concussion in sport: the 4th International Conference on Concussion in Sport, Zurich, November 2012. J Ath Train 2013;48(4):554–575.
24. McCrory P, Johnston K, Meeuwisse W, et al. Summary and agreement statement of the 2nd International Conference on Concussion in Sport, Prague 2004. Br J Sport Med 2005;39(4):196–204.
25. McCrory P, Meeuwisse W, Johnston K, et al. Consensus statement on concussion in sport—the 3rd International Conference on Concussion in Sport, held in Zurich, November 2008. J Clin Neurosci 2009;16(6):755–763.
26. Iverson GL, Lovell MR, Smith SS. Does brief loss of consciousness affect cognitive functioning after mild head injury? Arch Clin Neuropsychol 2000;15(7):643–648.
27. Lovell MR, Iverson GL, Collins MW, et al. Does loss of consciousness predict neuropsychological decrements after concussion? Clin J Sport Med 1999;9(4):193–198.
28. Guskiewicz KM, Ross SE, Marshall SW. Postural stability and neuropsychological deficits after concussion in collegiate athletes. J Athl Train 2001;36(3):263–273.
29. Hinton-Bayre AD, Geffen G. Severity of sports-related concussion and neuropsychological test performance. Neurology 2002;59(7):1068–1070.
30. Teel EF, Marshall SW, Shankar V, et al. Predicting recovery patterns after sport-related concussion. J Athl Train 2017;52(3):288–298.
31. Makdissi M, Darby D, Maruff P, et al. Natural history of concussion in sport: markers of severity and implications for management. Am J Sports Med 2010;38(3):464–471.
32. Meehan WP 3rd, Mannix RC, Stracciolini A, et al. Symptom severity predicts prolonged recovery after sport-related concussion, but age and amnesia do not. J Pediatr 2013;163(3):721–725.
33. Zuckerman SL, Yengo-Kahn AM, Buckley TA, et al. Predictors of post-concussion syndrome in collegiate student-athletes. Neurosurg Focus 2016;40(4):E13.
34. Lau BC, Kontos AP, Collins MW, et al. Which on-field signs/symptoms predict protracted recovery from sport-related concussion among high school football players? Am J Sports Med 2011;39(11):2311–2318.
35. Erlanger D, Kaushik T, Cantu R, et al. Symptom-based assessment of the severity of a concussion. J Neurosurg 2003;98(3):477–484.
36. Howell DR, Potter MN, Kirkwood MW, et al. Clinical predictors of symptom resolution for children and adolescents with sport-related concussion. J Neurosurg Pediatr 2019;24(1):54–61.
37. Kontos AP, Elbin RJ, Sufrinko A, et al. Recovery following sport-related concussion: integrating pre-and postinjury factors into multidisciplinary care. J Head Trauma Rehab 2019;34(6):394–401.
38. Jildeh TR, Okoroha KR, Denha E, et al. Return to sport following adolescent concussion: epidemiologic findings from a high school population. Orthopedics, 2020;43(4):e306–e310.
39. Bartsch AJ, Hedin D, Alberts J, et al. High energy side and rear American football head impacts cause obvious performance decrement on video. Ann Biomed Eng 2020:48(11):2667–2677.
40. Zuckerman SL, Elbin RJ, Sills AK, et al. Concussions in the National Football League: the evolution of video review for assessing the frequency and reliability of visible signs. Phys Sportsmed 2020;48(4):424–429.
41. Elbin RJ, Zuckerman SL, Sills AK, et al. Sensitivity and specificity of on-field visible signs of concussion in the National Football League. Neurosurgery, 2020;87(3):530–537.
42. Lessley DJ, Kent RW, Funk JR, et al. Video analysis of reported concussion events in the National Football League during the 2015-2016 and 2016-2017 seasons. Am J Sports Med 2018;46(14):3502–3510.
43. Austin AV, Sasser P, Tanabe K, et al. Video analysis of concussion exposures in a National Collegiate Athletic Association Division I football team. Orthop J Sports Med 2020;8(2):2325967120903294.
44. Davis G, Makdissi M. Use of video to facilitate sideline concussion diagnosis and management decision-making. J Sci Med Sport 2016;19(11):898–902.
45. Makdissi M, Davis G. The reliability and validity of video analysis for the assessment of the clinical signs of concussion in Australian football. J Sci Med Sport 2016;19(10):859–863.
46. Makdissi M, Davis G. Using video analysis for concussion surveillance in Australian football. J Sci Med Sport 2016;19(12):958–963.
47. McCrory PR, Berkovic SF. Video analysis of acute motor and convulsive manifestations in sport-related concussion. Neurology 2000;54(7):1488–1491.
48. Hutchison MG, Comper P, Meeuwisse WH, et al. A systematic video analysis of National Hockey League (NHL) concussions, part I: who, when, where and what? Br J Sports Med 2015;49(8):547–551.
49. Hutchison MG, Comper P, Meeuwisse WH, et al. A systematic video analysis of National Hockey League (NHL) concussions, part II: how concussions occur in the NHL. Br J Sports Med, 2015;49(8):552–555.
50. Echemendia, RJ, Bruce JM, Meeuwisse W, et al. Can visible signs predict concussion diagnosis in the National Hockey League? Br J Sports Med 2018;52(17):1149–1154.
51. Kohler R, Makdissi M, McDonald W, et al. A preliminary video review of in-game head injury incidents (HII) and use of the head injury assessment (HIA) from the 2015 super rugby season. Br J Sport Med 2017;51(15):A78.73–A78.A79.
52. Tucker R, Raftery M, Fuller GW, et al. A video analysis of head injuries satisfying the criteria for a head injury assessment in professional Rugby Union: a prospective cohort study. Br J Sports Med 2017;51(15):1147–1151.
53. Gardner AJ, Iverson GL, Quinn TN, et al. A preliminary video analysis of concussion in the National Rugby League. Brain Inj, 2015;29(10):1182–1185.
54. Gardner AJ, Iverson GL, Stanwell P, et al. A video analysis of use of the new 'Concussion Interchange Rule' in the National Rugby League. Int J Sport Med 2016;37(4):267–273.
55. Gardner AJ, Kohler RM, Levi CR et al. Usefulness of video review of possible concussions in National Youth Rugby League. Int J Sports Med 2017;38(1):71–75.
56. Gardner AJ, Howell DR, Levi CR et al. Evidence of concussion signs in National Rugby League match play: a video review and validation study. Sport Med Open 2017;3(1):29.
57. Gardner AJ, Howell DR, Iverson GL. A video review of multiple concussion signs in National Rugby League match play. Sports Med Open 2018;4(1):5.
58. Iverson GL, Van Patten R, Gardner AJ. Examining whether onfield motor incoordination is associated with worse performance on the SCAT5 and slower clinical recovery following concussion. Frontiers in Neurol 2021;11:620872.
59. Kindschi K, Higgins M, Hillman A, et al. Video analysis of high-magnitude head impacts in men's collegiate lacrosse. BMJ Open Sport Exerc Med 2017;3(1):e000165.
60. Lincoln AE, Caswell SV, Almquist JL, et al. Video incident analysis of concussions in boys' high school lacrosse. Am J Sports Med 2013;41(4):756–761.
61. Koh JO, Watkinson EJ, Yoon YJ. Video analysis of head blows leading to concussion in competition Taekwondo. Brain Inj 2004;18(12):1287–1296.
62. Davis GA, Makdissi M, Bloomfield P, et al. International consensus definitions of video signs of concussion in professional sports. Br J Sports Med 2019;53(20):1264–1267.
63. Echemendia RJ, Meeuwisse W, McCrory P, et al. The Sport Concussion Assessment Tool 5th Edition (SCAT5): background and rationale. Br J Sport Med 2017;51(11):848–850.
64. Gardner AJ, Kohler R, McDonald W, et al. The use of sideline video review to facilitate management decisions following head trauma in super rugby. Sports Med Open 2018;4(1):20.
65. Gardner AJ, Wojtowicz M, Terry DP, et al. Video and clinical screening of National Rugby League players suspected of sustaining concussion. Brain Inj 2017;31(13-14):1918–1924.
66. Yengo-Kahn AM, Hale AT, Zalneraitis BH, et al. The Sport Concussion Assessment Tool: a systematic review. Neurosurg Focus 2016;40(4):E6.
67. Katz BP, Kudela M, Harezlak J, et al. Baseline performance of NCAA athletes on a concussion assessment battery: a report from the CARE Consortium. Sports Med 2018;48(8):1971–1985.
68. Fuller GW, Raftery M. Sport Concussion Assessment Tool: Fifth Edition Normative Reference Values for professional Rugby Union players. Clin J Sport Med 2020;30(5):e150–e153.
69. Fuller GW, Tucker R, Starling L, et al. The performance of the World Rugby Head Injury Assessment Screening Tool: a diagnostic accuracy study. Sports Med Open 2020;6(1):2.

Downloaded by Massachusetts Institute of Technology from www.liebertpub.com at 11/21/23. For personal use only.

70. Cochrane GD, Owen M, Ackerson JD, et al. Exploration of US men's professional sport organization concussion policies. Phys Sportsmed 2017;45(2):178–183.
71. McCrea M, Kelly JP, Randolph C, et al. Standardized assessment of concussion (SAC): on-site mental status evaluation of the athlete. J Head Trauma Rehabil 1998;13(2):27–35.
72. Riemann BL, Guskiewicz KM. Effects of mild head injury on postural stability as measured through clinical balance testing. J Athl Train 2000;35(1):19–25.
73. Cohen J. Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Lawrence Erlbaum Associates, Publishers: Hillsdale, NJ; 1988.
74. Cohen J. A power primer. Psychol Bull 1992;112(1):155–159.
75. Lenhard W, Lenhard A. Computation of effect sizes. Psychometrica; 2016. doi: 10.13140/RG.2.2.17823.92329
76. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. J Royal Stat Soc: Series B (Methodological) 1995;57(1):289–300.
77. Meehan WP 3rd, d'Hemecourt P, Comstock RD. High school concussions in the 2008-2009 academic year: mechanism, symptoms, and management. Am J Sports Med 2010;38(12):2405–2409.
78. Pellman EJ, Powell JW, Viano DC, et al. Concussion in professional football: epidemiological features of game injuries and review of the literature—part 3. Neurosurgery 2004;54(1):81–94.
79. Cassoudesalle H, Laborde B, Orhant E, et al. Video analysis of concussion mechanisms and immediate management in French men's professional football (soccer) from 2015 to 2019. Scand J Med Sci Sports 2021;31(2):465–472.
80. Gil C, Jacota M, Caudron Y, et al. How is video efficient to diagnose sport-related concussion? a cross-sectional study in the French TOP14 Rugby Championship. Clin J Sport Med 2021;32(3):e261–e267.
81. Iverson GL, Gardner AJ, Terry DP, et al. Predictors of clinical recovery from concussion: a systematic review. Br J Sports Med 2017;51(12):941–948.
82. Dougan BK, Horswill MS, Geffen GM. Do injury characteristics predict the severity of acute neuropsychological deficits following sports-related concussion? A meta-analysis. J Int Neuropsychol Soc 2014;20(1):81–87.
83. Gardner AJ. Reliability of using the proposed international consensus video signs of potential concussion for National Rugby League head impact events. Neurosurgery 2021;88(3):538–543.

Downloaded by Massachusetts Institute of Technology from www.liebertpub.com at 11/21/23. For personal use only.